

Statistics Portfolio

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1 Statistics Portfolio

1.1 About

Valdinar Pires Student at the University of Campinas (UNICAMP), with an interest in statistical analysis, data visualization, and statistical modeling.

This portfolio brings together projects developed throughout my academic training, with an emphasis on applying statistical concepts to real-world and historical datasets.

1.2 Projects

The projects presented here explore different statistical methods and approaches, combining theoretical foundations with practical analysis in R and other data-related tools.

1.2.1 Current projects

- **Semmelweis Handwashing Study** Analysis of historical data from the Vienna General Hospital, investigating the relationship between handwashing and maternal mortality.

More projects will be added as my academic portfolio develops.

1.3 Tools

The main tools used across the projects include:

- R
- Quarto
- Git and GitHub

Additional tools and programming languages will be incorporated as the portfolio expands.

2 About Me

I am a Statistics student at the University of Campinas (UNICAMP), interested in understanding how statistical methods can be used to extract information from data and support quantitative reasoning.

My academic training combines statistical theory, mathematical foundations, and practical data analysis. Through coursework and independent projects, I have been developing skills in statistical programming, data visualization, modeling, and statistical inference.

2.1 Academic Interests

My main interests include:

- Statistical inference
- Sampling and survey methodology
- Regression and statistical modeling
- Experimental design
- Exploratory and descriptive data analysis
- Data visualization

2.2 Working with Data

I am particularly interested in the process of moving from raw data to meaningful statistical conclusions: understanding the structure of a dataset, selecting appropriate methods, evaluating assumptions, and communicating results clearly.

This portfolio serves as a record of that development, bringing together projects completed during my academic training and documenting the methods used in each analysis.

3 Projects

This section presents selected statistical projects developed throughout my academic training.

Each project focuses on a specific dataset, statistical question, or methodological problem, combining statistical reasoning with reproducible data analysis.

3.1 Semmelweis Handwashing Study

Statistical analysis of handwashing and maternal mortality

This project analyzes historical data from the Vienna General Hospital between 1841 and 1846 to investigate the relationship between handwashing practices and maternal mortality.

The analysis includes:

- Comparison of mortality between two clinics;
- Descriptive statistics and data visualization;
- Comparison of mortality before and after the introduction of handwashing;
- Comparison of mean, median, and mode;
- Hypothesis testing.

Tools: R · tidyverse · ggplot2 · Quarto

[View project →](#)

3.2 More Projects

Additional projects will be added as my academic portfolio develops, covering topics such as sampling, regression, experimental design, probability, statistical inference, and data analysis.

References